HealthConnect Week 8

1. Executive Summary

Week 8 converts the validated HealthConnect analytics work from Weeks 5–7 into a final decision-support package. The focus remains the HealthConnect appointment-attendance problem: reducing missed appointments, improving attendance, making better use of appointment slots, and supporting patients administratively.

The final analytics component is the refined HealthConnect Power BI dashboard and its supporting business-insight documentation. Week 7 testing validated the core dashboard measures and identified the remaining integration needs. The final package therefore emphasizes validated KPIs, attendance/no-show patterns, actionable business insights, limitations, cross-track contribution, and presentation readiness.

Key validated figures carried into the final package are: 1,696 patients; 48.46% no-show rate; 46.28% attendance rate; and 5.26% cancellation rate. Testing also identified important patterns including a 60.49% no-show rate for appointments 31+ days out, 67.88% for 31+ days out among patients with a previous no-show, and 57.76% for appointments involving distances of 21+ km.

2. Week 7 → Week 8 Transition

Requirement	Week 8 response
Main Week 7 output	Analytics Testing & Refinement Report based on the Week 6 HealthConnect dashboard, including KPI/source reconciliation, visual and filter testing, issue identification, refinement, re-testing, findings, recommendations and limitations.
Most important testing result	The core dashboard figures were validated: 48.46% no-show rate, 46.28% attendance rate, 5.26% cancellation rate and 1,696 total patients.
Major issue discovered	The dashboard still required manual Power BI slicer click-through/testing for interaction validation, while direct Data Science re-testing/model-artifact evidence remained an integration need.
What was refined	The analytics work was refined around validated KPIs, clearer business interpretation, tested attendance/no-show patterns, updated recommendations and explicit limitations.
Component now considered ready	The core Data Analytics dashboard/decision-support component and its validated analytical findings are ready to contribute to the final HealthConnect integration.
What still needs integration	Final exchange of validated analytics findings with the Data Science track, final presentation

	sequencing, and confirmation of how the analytics output is used in the multidisciplinary walkthrough.
Tracks involved	Data Science for validated analytical findings that support model interpretation; Project Management for final business recommendations, project decisions, presentation and readiness coordination; other tracks as required by the final HC-POD walkthrough.
Presentation demonstration	The final dashboard, validated KPIs, major no-show/attendance patterns, business implications, recommendations, limitations, and how analytics feeds the wider HealthConnect solution.

3. Final Integration Readiness

Requirement	Final response
What is my final component?	HealthConnect Final Analytics & Decision Support Component: the refined Power BI dashboard plus validated KPIs, findings, business insights, recommendations, limitations and presentation material.
What problem does it address?	It helps HealthConnect understand missed appointments and attendance patterns so that appointment scheduling, patient support and administrative decisions can be better informed.
Current status	Core analytics component ready for final integration and presentation.
Week 7 improvement	Testing and refinement validated the main KPIs, checked dashboard figures and strengthened the interpretation of attendance/no-show patterns and business recommendations.
Dependencies	Data Science can use validated analytics findings for model interpretation; Project Management needs final insights for business recommendations and presentation coordination.
Tracks depending on this output	Primarily Data Science and Project Management, with the final HC-POD using the analytics output as the evidence base for the end-to-end solution.
Output provided	Final dashboard/report, KPI set, validated findings, key visualizations, business insights, recommendations, limitations, executive summary and presentation material.
Output needed from another track	Final confirmation of how Data Science incorporates the validated analytics findings into the final model interpretation; final integration/presentation coordination from Project Management.
Evidence of readiness	Week 7 testing results, validated KPI figures, documented patterns, refinements and explicit remaining limitations.
Remaining limitations	Some interactive dashboard behavior still requires manual Power BI validation, and the final Data Science model artifact/re-testing evidence is an

	integration dependency rather than an analytics result.

4. Final KPIs

KPI	Validated Value	Decision-support meaning
Total Patients	1,696	Defines the patient/appointment population represented in the final dashboard analysis.
No-Show Rate	48.46%	Shows the scale of missed appointments and the need for attendance-focused intervention.
Attendance Rate	46.28%	Shows the proportion of appointments recorded as attended.
Cancellation Rate	5.26%	Separates cancellations from no-shows and prevents cancellations from being treated as automatic no-shows.

5. Validated Findings & Key Patterns

- Appointments scheduled 31+ days out showed a 60.49% no-show rate.
- Among appointments scheduled 31+ days out for patients with a previous no-show, the no-show rate was 67.88%.
- Appointments involving distances of 21+ km showed a 57.76% no-show rate.
- The overall no-show rate of 48.46% indicates that missed appointments are a major operational issue within the analyzed HealthConnect dataset
- Cancellation should remain a distinct category because it represents a different appointment outcome from a no-show.

6. Key Visualizations for the Final Dashboard

- Executive KPI cards: Total Patients, No-Show Rate, Attendance Rate and Cancellation Rate.
- Appointment lead-time analysis highlighting the 31+ day group.
- Previous no-show analysis to identify patients with a prior attendance-risk signal.
- Distance analysis highlighting the 21+ km group.
- Attendance/no-show distribution visualizations that allow stakeholders to see the scale of the problem.
- Interactive slicers/filters used to explore relevant patient and appointment dimensions, subject to the documented manual interaction-validation limitation.

7. Business Insights

Insight	Business interpretation
High overall missed-appointment level	Nearly half of the analyzed appointments were no-shows, making attendance management a central operational concern.

Longer appointment lead time is associated with higher observed no-show levels	The 31+ day group has a 60.49% no-show rate, indicating that appointments scheduled far in advance may require additional attendance-support attention.
Previous no-show is an important observed risk signal	The 67.88% no-show rate in the 31+ day + previous-no-show group identifies a particularly high observed-risk segment for administrative follow-up.
Distance is associated with attendance behavior in the analyzed data	The 21+ km group has a 57.76% no-show rate, suggesting that travel distance may be relevant when designing patient-support approaches.
Cancellations require separate handling	The 5.26% cancellation rate should not be merged into no-show reporting because the two outcomes represent different administrative events.

8. Actionable Recommendations

- Use the dashboard to monitor no-show and attendance KPIs as recurring operational measures.
- Give additional administrative attention to appointments scheduled 31+ days in advance, especially where a previous no-show is recorded.
- Consider targeted appointment reminders or confirmation workflows for higher-observed-risk groups, subject to HealthConnect's approved operational policies.
- Use distance-related patterns to inform patient-support and appointment-management discussions without treating distance as proof of individual patient behavior.
- Keep cancellation reporting separate from no-show reporting so operational teams can respond appropriately to each outcome.
- Use the validated analytics findings as supporting evidence for the Data Science track's interpretation of any final predictive component.

9. Decision Support for HealthConnect

The analytics component supports HealthConnect decisions by converting appointment records into measurable operational indicators and identifiable patterns. The dashboard can help stakeholders monitor the scale of missed appointments, identify segments with higher observed no-show levels, and prioritize administrative attention. The analytics should be used as decision support rather than as a substitute for clinical judgment or as proof of causation.

10. Cross-Track Contribution & Final Integration

Track	Dependency / exchange	Final integration activity	Expected change/value
Data Science	Analytics provides validated no-show/attendance patterns and KPI definitions.	Use validated findings to support final model interpretation and business explanation.	The predictive component is interpreted using consistent analytical definitions and observed patterns.
Project Management	Analytics provides final insights, recommendations and limitations.	Incorporate analytics evidence into final business recommendations,	The project story connects the data evidence to the wider HealthConnect solution.

		presentation sequence and readiness assessment.	
HC-POD / All Tracks	Analytics provides the final decision-support layer.	Use the dashboard and findings during the end-to-end integration walkthrough.	The final solution demonstrates how analytical evidence connects with predictive, engineering, GenAI and coordination components.

11. Analytical Limitations

- The dataset represents the approved HealthConnect project resource and should not automatically be treated as representative of all healthcare settings.
- Observed associations should not be presented as causal relationships. For example, a higher no-show rate among long-lead-time or long-distance appointments does not by itself prove that lead time or distance causes non-attendance.
- The dashboard's interactive behavior still requires manual Power BI click-through validation for some slicer/filter scenarios.
- The final Data Science model artifact and direct model re-testing remain cross-track integration dependencies rather than completed Data Analytics outputs.
- The analytics component supports administrative decision-making; it does not provide clinical diagnosis or medical advice.
- Any future operational intervention should be evaluated against updated attendance data to determine whether it improves outcomes.

12. HC-POD Final Integration Walkthrough — Analytics Role

Walkthrough stage	Data Analytics contribution
1. Problem	Quantifies the attendance/no-show problem.
2. Data	Explains the approved HealthConnect appointment data used for analysis.
3. Analytics	Shows the final KPIs, validated patterns and dashboard.
4. Predictive Component	Provides validated analytical findings that can support model interpretation.
5. ML Engineering	Provides the analytics context required to understand how the final technical workflow uses the predictive component.
6. GenAI	Provides business/analytical context for the administrative-support story; the GenAI component remains responsible for its own approved knowledge-base scope and safety behavior.
7. Project Management	Provides final insights, recommendations, limitations and evidence for project decisions and presentation.
8. Final Solution	Positions the dashboard as the decision-support/evidence layer within the multidisciplinary solution.

9. Business Value	Shows where HealthConnect may focus attendance-management and administrative-support attention.
10. Limitations	Clearly communicates dashboard validation and cross-track dependencies that remain.